

1. 30p. Consider the following LP problem and its optimal simplex table. Here, x and y represent the number of units to be produced for product 1 and product 2, respectively, while the slack variable added to the constraint i is denoted by s_i for $i = 1, 2$.

$$\begin{aligned} \max Z &= 3x + 2y \\ \text{s.t. } x + 2y &\leq 12 \\ 2x + y &\leq 15 \\ x, y &\geq 0. \end{aligned}$$

(Resource A) 1
(Resource B) 2

		d_1	d_2			
Basic	x	y	s_1	s_2		RHS
Z	0	0	$1/3$	$4/3$		24
y	0	1	$2/3$	$-1/3$		3
x	1	0	$-1/3$	$2/3$		6

$$z = 24 + \frac{1}{3}D_1 + \frac{4}{3}D_2$$

$$y = 3 + \frac{2}{3}D_1 - \frac{1}{3}D_2$$

$$x = 6 - \frac{1}{3}D_1 + \frac{2}{3}D_2$$

(i) What is the dual price for the resource A constraint?

- A. $-4/3$ B. $-1/3$ C. $1/3$ D. $4/3$ E. Postoptimality is required.

(ii) When the amount of Resource B is decreased from 15 to 12, what will be the change amount in the objective function value?

- A. -4 B. -1 C. 1 D. 4 E. Postoptimality is required.

(iii) Change the amounts of Resources A and B to 33 and 27, respectively. What would be the optimal objective function value?

- A. -4 B. -1 C. 1 D. 4 E. Postoptimality is required.

$$RHS_1 = 33 \Rightarrow D_1 = 21$$

$$RHS_2 = 27 \Rightarrow D_2 = 12$$

$$y = 3 + \frac{2}{3} \cdot 21 - \frac{1}{3} \cdot 12 = 3 + 14 - 4 = 13 \quad \checkmark$$

$$x = 6 - \frac{1}{3} \cdot 21 + \frac{2}{3} \cdot 12 = 6 - 7 + 8 = 7 \quad \checkmark$$

z can be calculated

$$z = 24 + \frac{1}{3} \cdot 21 + \frac{4}{3} \cdot 12 = 24 + 7 + 16 = 47$$

(iv) What is the feasibility range of Resource A?

- A. $[-9/2, 18]$

- B. $[6, 24]$

- C. $[15/2, 30]$

(v) What is the optimality range of the cost of x (i.e. c_1)?

- A. $[-2, 1]$

- B. $[1, 4]$

- C. $[-1/5, 4]$

- D. $[3/2, 6]$

- E. None

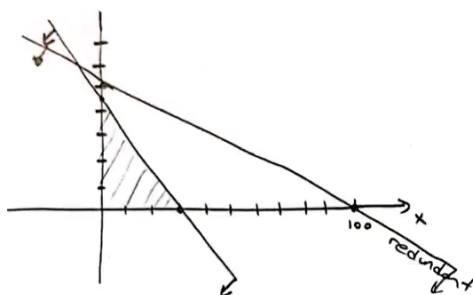
(vi) Change the costs of variables x and y to 6. What would be the optimal objective function value?

- A. 24 B. 34 C. 54 D. 78 E. Postoptimality is required.

2. 10p. One unit of product P_1 requires 3 kg of resource R_1 and 1 kg of resource R_2 . One unit of product P_2 requires 2 kg of resource R_1 and 2 kg of resource R_2 . The profits per unit by selling product P_1 and P_2 are 2 and 3, respectively. The manufacturer has 90 kg of resource R_1 and 100 kg of resource R_2 . The manufacturer can make a maximum profit of _____

Hint: You may use the graphical solution method. a. 100 b. 135 c. 150 d. 200 e. None

$$\begin{aligned} \max z &= 2x + 3y \\ 3x + 2y &\leq 90 \quad \dots (1) \\ x + 2y &\leq 100 \quad \dots (2) \\ x, y &\geq 0 \end{aligned}$$



$$(0, 45) \Rightarrow z = 135$$

$$(30, 0) \Rightarrow z = 60$$

$$x^* = 0, y^* = 45$$

$$z^* = 135$$

7. Solve the given problem by graphical method (10p), algebraic method (by enumeration) (10p), and Simplex

method (15p) and write the optimal solutions and the obj. func. value.

$$\min Z = x_1 + 2x_2$$

$$\text{s.t. } x_1 + x_2 \geq 2$$

$$3x_1 + 4x_2 \leq 8$$

$$x_1, x_2 \geq 0$$

$$D_2 = 0 \Rightarrow 3 + \frac{2}{3}D_1 \geq 0 \quad 6 - \frac{1}{3}D_1 \geq 0$$

$$D_1 \geq -\frac{9}{2} \quad 18 \geq D_1$$

$$-\frac{9}{2} \leq D_1 \leq 18 \Rightarrow \frac{15}{2} \leq 12 + D_1 = \text{RHS}_1 \leq 30$$

$$12 + \left(\frac{-9}{2}\right)$$

$$\left. \begin{aligned} D_1 = 0 \Rightarrow 3 - \frac{1}{3}D_2 \geq 0 &\Rightarrow 9 \geq D_2 \\ 6 + \frac{2}{3}D_2 \geq 0 &\Rightarrow D_2 \geq -9 \end{aligned} \right\} 6 \leq \underbrace{15 + D_2}_{\text{RHS}_2} \leq 24$$

$$\text{RHS}_1 = 33 \Rightarrow D_1 = 21$$

$$\text{RHS}_2 = 27 \Rightarrow D_2 = 12$$

3. 15p. Consider the following linear program

$$\min z = 3x_1 + 2x_2 - 3x_3 + 4x_4$$

$$\text{s.t. } x_1 - 2x_2 + 4x_3 + 4x_4 \geq 3$$

$$x_2 + 3x_3 + 4x_4 \geq -5$$

$$2x_1 - 3x_2 - 7x_3 - 4x_4 = 2$$

$$x_1, x_3, x_4 \geq 0.$$

i) Which of the following statement(s) is/are correct?

i. The dual variable corresponding to the first constraint can take negative values.

ii. The dual variable corresponding to the second constraint can take negative values.

iii. The dual variable corresponding to the third constraint can take negative values.

A. Only i B. Only ii C. i and ii **D. ii and iii** E. i, ii and iii

$$\max z = 6x_1 + 8x_2$$

$$x_1 + x_2 + s_1 = 40$$

$$2x_1 + 3x_2 - s_2 + r_1 = 15$$

$$x_1 - x_2 + r_2 = 5$$

$$x_1, x_2, s_1, s_2, r_1, r_2 \geq 0$$

The basis: $\begin{bmatrix} s_2 \\ x_2 \\ x_1 \end{bmatrix} \Rightarrow B = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 3 & 2 \\ 0 & -1 & 1 \end{bmatrix}$ original coefficient matrix. starting basis is $(s_1, r_1, r_2) \Rightarrow$

$$\Rightarrow B^{-1} = \begin{bmatrix} 5/2 & -1 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 0 & 1/2 \end{bmatrix} \text{ since the given table is optimal.}$$

ii. Which can be the dual objective function of the given problem?

A Maximize $w = 3y_1 + 5y_2 + 2y_3$

B. Maximize $w = 3y_1 - 5y_2 + 2y_3 + 4y_4$

C. Minimize $w = 3y_1 - 5y_2 + 2y_3$

D. Maximize $w = 3y_1 - 5y_2$

E. None.

iii. In the dual problem, which constraint corresponds to the primal variable x_1 ?

A. $y_1 - y_3 \leq 3$

B $y_1 + 2y_3 \leq 3$

C. $y_1 - 2y_3 \leq 3$

D. $y_1 + 2y_3 \geq 3$

E. None.

4. 15p. Consider the following problem as the primal problem and its simplex tableau at optimality.

$\max z= 6x_1 + 8x_2$									
s.t.	$x_1 + x_2 \leq 10$								
	$2x_1 + 3x_2 \geq 15$								
	$x_1 - x_2 = 5$								
	$x_1, x_2 \geq 0$								
		Basis	x_1	x_2	s_1	s_2	R_1	R_2	Solution
		z-row	0	0	7	0	M	-1+M	65
		s_2	0	0	5/2	1	-1	-1/2	15/2
		x_2	0	1	1/2	0	0	-1/2	5/2
		x_1	1	0	1/2	0	0	1/2	15/2

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i. What is the inverse of this basis matrix (B^{-1})?

- A. $\begin{bmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{bmatrix}$
- ☒ C. $\begin{bmatrix} 5/2 & -1 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 0 & 1/2 \end{bmatrix}$
- B. $\begin{bmatrix} 5/2 & 1 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 0 & 1/2 \end{bmatrix}$
- D. $\begin{bmatrix} 5/2 & -1/2 \\ 1/2 & -1/2 \end{bmatrix}$
- E. None of the above.

ii. What are the optimal values of the dual variables y_1, y_2, y_3 (associated with the primal constraints $\leq, \geq, =$, respectively)?

- A. $y_1 = 7, y_2 = M, y_3 = -1 + M$
- B. $y_1 = -1, y_2 = 7, y_3 = 0$
- ☒ C. $y_1 = 7, y_2 = 0, y_3 = -1$
- D. $y_1 = -1, y_2 = 0, y_3 = 7$
- E. None of the above.

(s_1, R_1, R_2) correspond to (y_1, y_2, y_3)

$$y_1 = 7 + 0 = 7$$

$$y_2 = M - M = 0$$

$$y_3 = -1 + M - M = -1$$

iii. During an intermediate simplex step in the given problem, the current primal objective function value is $z = 44$. Which of the following statements about the relationship between the primal and dual objective functions is correct, assuming the dual solution is feasible?

- A. The dual objective value w must equal $z = 44$.
- ☒ B. The dual objective value w must be greater than $z = 44$.
- C. The dual objective value w must be less than $z = 44$.
- D. The dual objective value w must be infeasible at this step.
- E. None of the above.

Given problem is maximization.

$$\begin{array}{l} \max z \\ z_0 \leq z_1 \leq z_2 \dots \\ z = 44 \end{array} \quad \begin{array}{l} \min w \\ z^* = w^* \leq \dots \leq w_2 \leq w_1 \leq w_0. \\ w > 44 \end{array}$$

5. 15p. Show that the set of all feasible solutions to an LP problem is a convex set.

Proof:

$$\begin{array}{l} \min z = CX \\ \text{s.t. } Ax = b \\ x \geq 0 \end{array}$$

we need to show that every convex combination of any two feasible solutions is also a feasible solution. Notice that, the theorem is true, if the set of solutions has only one element.

Assume there are at least two solutions x^1 and x^2 .

We have

$$Ax^1 = b \text{ for } x^1 \geq 0 \text{ and } Ax^2 = b \text{ for } x^2 \geq 0.$$

For $0 \leq \alpha \leq 1$, let $x = \alpha x^1 + (1-\alpha)x^2$ be any convex combination of x^1 and x^2 . Since all elements of the vector x are nonnegative, $x \geq 0$. [$\alpha \geq 0, x^1 \geq 0, (1-\alpha) \geq 0, x^2 \geq 0$].

$$Ax = b \Rightarrow Ax = A(\alpha x^1 + (1-\alpha)x^2) = \alpha Ax^1 + (1-\alpha)Ax^2 = \alpha b + (1-\alpha)b = b //$$

$Ax = b$ and $x \geq 0$, thus x is feasible.

So, every convex combination of any two feasible solutions is also a feasible solution. Thus, the feasible region is convex.

6. 10p. a. Take the dual of the problem below.

b. 5p. Write the starting basic feasible solution of the dual.

$$\min Z = 6x_1 + 7x_2 + 4x_3$$

$$\text{s.t. } 2x_1 - x_2 + x_3 \leq 5$$

$$3x_2 + x_3 \geq -4$$

$$x_1 - 2x_2 - 5x_3 = 3$$

$$x_1, x_3 \geq 0$$

$$x_2 \text{ free} \Rightarrow x_2 = x_2' - x_2''$$

$$\min Z = 6x_1 + 7x_2' - 7x_2'' + 4x_3 + MR_L$$

$$\text{s.t. } 2x_1 - x_2' + x_2'' + x_3 + S_1 = 5 \quad y_1$$

$$-3x_2' + 3x_2'' - x_3 + S_2 = 4 \quad y_2$$

$$x_1 - 2x_2' + 2x_2'' - 5x_3 + R_1 = 3 \quad y_3$$

$$x_1, x_2', x_2'', x_3, S_1, S_2, R_1 \geq 0$$

$$\max W = 5y_1 + 4y_2 + 3y_3$$

$$2y_1 + y_3 \leq 6$$

$$-y_1 - 3y_2 - 2y_3 \leq 7$$

$$y_1 + 3y_2 + 2y_3 \leq -7$$

$$y_1 - y_2 - 5y_3 \leq 4$$

$$-y_1 - 3y_2 - 2y_3 = 7$$

$$\underbrace{-y_1}_{y_1'} \geq 0 \quad \underbrace{-y_2}_{y_2'} \geq 0 \quad y_3 = y_3' - y_3''$$

$$\left. \begin{array}{l} \max W = -5y_1' - 4y_2' + 3y_3' - 3y_3'' - MR_L \\ -2y_1' + y_3' - y_3'' + S_3 = 6 \\ y_1' + 3y_2' - 2y_3' + 2y_3'' + R_2 = 7 \\ -y_1' + y_2' - 5y_3' + 5y_3'' + S_4 = 4 \end{array} \right\} \begin{array}{l} \text{starting b.s.} \\ (S_3, R_2, S_4) = (6, 7, 4) \\ \text{others} = 0. \end{array}$$

$$\begin{array}{rcl}
 y_1 & \leq 0 & \\
 y_2 & \leq 0 & \\
 y_3 & \leq M & \\
 y_1, y_2, y_3 \text{ free.} & &
 \end{array}
 \left. \vphantom{\begin{array}{rcl} y_1 \\ y_2 \\ y_3 \end{array}} \right\}
 \begin{array}{l}
 y_1 \leq 0 \\
 y_2 \leq 0 \\
 y_3 \text{ is free.}
 \end{array}$$

